

Exciton–Photonics: From Fundamental Science to Applications

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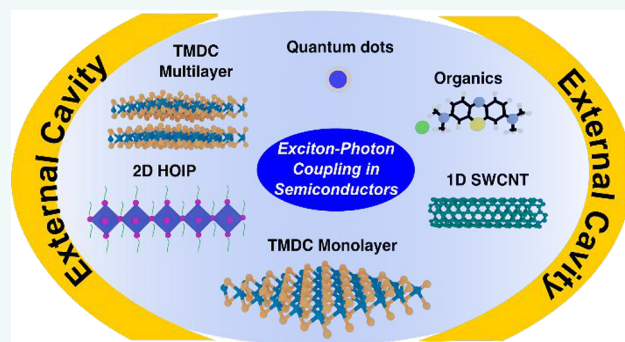
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ABSTRACT: Semiconductors in all dimensionalities ranging from 0D quantum dots and molecules to 3D bulk crystals support bound electron–hole pair quasiparticles termed excitons. Over the past two decades, the emergence of a variety of low-dimensional semiconductors that support excitons combined with advances in nano-optics and photonics has burgeoned an advanced area of research that focuses on engineering, imaging, and modulating the coupling between excitons and photons, resulting in the formation of hybrid quasiparticles termed exciton–polaritons. This advanced area has the potential to bring about a paradigm shift in quantum optics, as well as classical optoelectronic devices. Here, we present a review on the coupling of light in excitonic semiconductors and previous investigations of the optical properties of these hybrid quasiparticles *via* both far-field and near-field imaging and spectroscopy techniques. Special emphasis is given to recent advances with critical evaluation of the bottlenecks that plague various materials toward practical device implementations including quantum light sources. Our review highlights a growing need for excitonic material development together with optical engineering and imaging techniques to harness the utility of excitons and their host materials for a variety of applications.

KEYWORDS: near-field spectroscopy, excitons, exciton–polaritons, plasmonics, plexcitons, nanophotonics, light–matter coupling, single-photon emitters, low-dimensional materials, electrically controlled emission



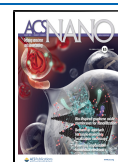
Semiconductors are the workhorse materials class of modern microelectronics and optoelectronics. While electronic devices are mostly reliant on Si, optoelectronic and photonic devices have been dominated by compound semiconductors. The operating principles of these devices rely on the interaction of light with the crystal medium. In most cases, the interaction is simple propagation of light or absorption of the photon that results in excitation of a bound electron from the valence band into a free electron in the conduction band, leaving behind a free hole. However, as the device dimensions shrink, the case of tight integration of optoelectronics and photonics with electronics becomes stronger. From this perspective, it is imperative to understand the interaction of light with semiconductors under strong physical confinement. This is a frontier area for basic as well as applied research in nanophotonics, and undoubtedly, a tremendous amount of work has been done and is being done with regards to light confinement in nanowires, nanorods, and quantum wells of Si as well as compound semiconductors. However, another approach is to study light–matter interactions in semiconductors that are naturally “low-

dimensional” or “quantum-confined”. The interaction of light with such crystals is fundamentally different since an above gap photon excites a tightly bound electron–hole pair called the exciton, as shown in Figure 1A. These tightly bound excited-state quasiparticles offer a different regime for light confinement in deep subwavelength dimensions. Low-dimensional semiconductors show strong excitonic resonances with high exciton binding energies, due to the weak screening of electron–hole pairs. Consequently, their optical density of states is localized to narrow spectral resonances, which makes them favorable for strong light–matter interaction. Adopting these materials in devices is expected to drive a transition from optoelectronic devices to opto-excitonic or exciton–photon devices where both light and exciton interact strongly in the

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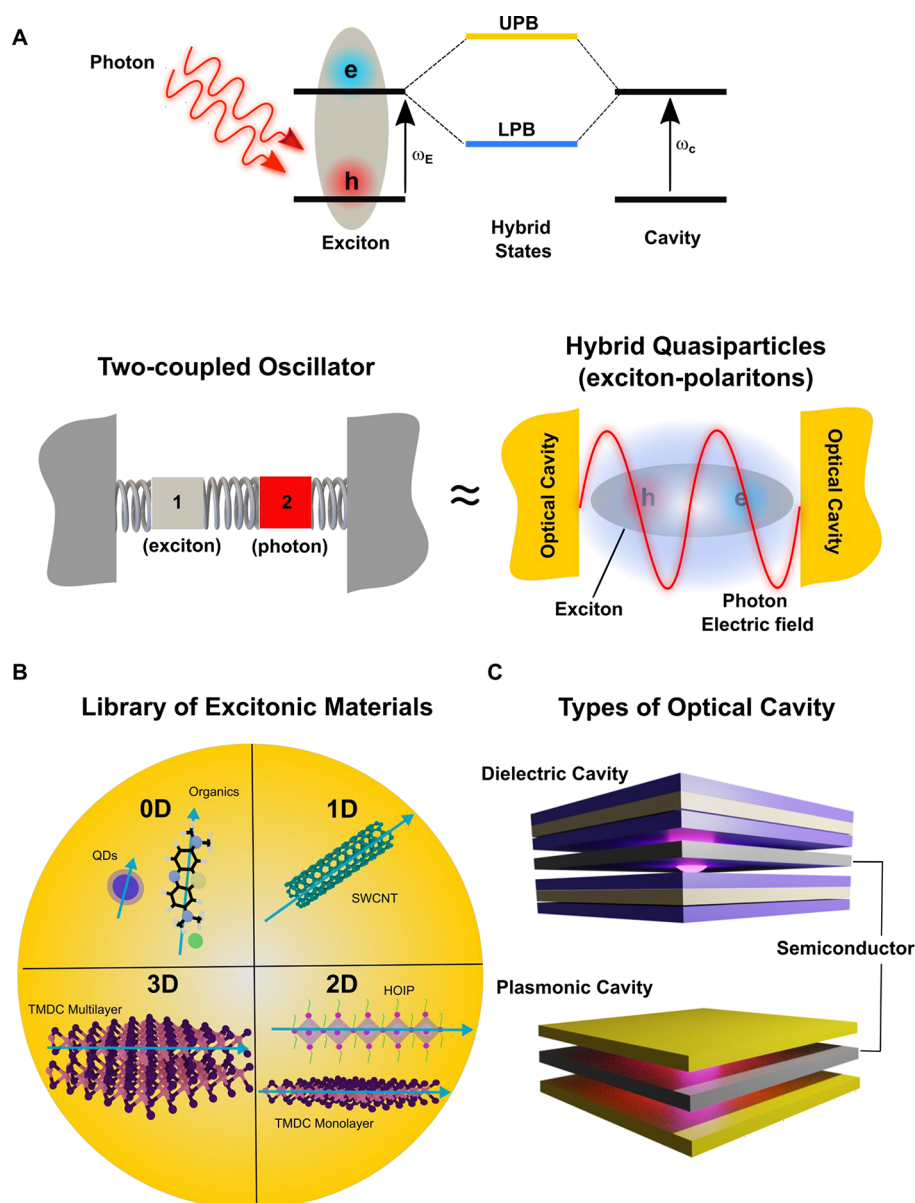


Figure 1. Light–matter coupling in the semiconductors. (A) Top panel: Exciton (bound electron–hole pair) formation upon absorption of the incident photons above the exciton resonance frequency (ω_E) of the semiconductor. Hybrid state [upper polariton branch (UPB) and lower polariton branch (LPB)] formation when the exciton resonantly interacts with a cavity photon or plasmon frequency (ω_c). Bottom panel: Schematic showing the spring-dashpot model for the two-coupled mechanical oscillator (left panel) as equivalent to exciton–photon coupling in an optical cavity (right panel). Formation of exciton–polaritons when the transition dipole moment of the exciton is aligned with the photon electric field inside an optical cavity. (B) Library of excitonic materials from 0D QDs and organic molecules; 1D SWCNTs; 2D hybrid organic–inorganic perovskites, transition metal dichalcogenide monolayer, and 3D TMDC multilayer. The blue arrow points to the transition dipole moment (μ) in the excitonic medium. (C) Schematic showing a dielectric and plasmonic cavity as an external cavity medium for different semiconductor materials placed inside them.

same or closely coupled transport media leading to devices with hybrid functionalities.

Achieving stable excitons at room temperature depends on the dielectric constant of the material that can influence the Coulomb interaction between electrons and holes. Further, confinement of the excitons leads to an increase in binding energy which scales with reduction in semiconductor dimensions. In general, 0D semiconductors possess exciton binding energies higher than those of their 3D counterparts due to their higher confinement. Beyond understanding the functioning of optoelectronic devices, the neutral and quasiparticle nature of excitons has been explored for

fundamental studies in solid-state physics and quantum optics. Excitons can be broadly classified into Frenkel and Wannier–Mott excitons based on the electron–hole binding energy, as explained in Box 1. Frenkel excitons are observed in organic molecules,¹ molecular assemblies (J-aggregates),² and π -conjugated organic chains³ owing to their low dielectric constant (and strong binding energy ~ 0.5 to 1 eV), facilitating stable excitons at room temperature. Furthermore, these Frenkel excitons have a small Bohr radius that are localized to the molecule with limited diffusion constant. On the other hand, Wannier–Mott excitons possess a large Bohr radius compared to their lattice constant, high diffusion constant, and

Box 1. Vocabulary

[Terminology in Excitonics]

Binding energy: The binding energy of 1 s exciton is the difference between electronic and optical band gap. Nano-materials usually exhibit high binding energy due to quantum confinement, *i.e.*, in a 2D semiconductor, ~ 100 meV, hence stable at room temperature.

Decay Time: Time dimension quantity describing how long exciton lasts without recombination. Eventually the exciton decays by either radiative or nonradiative recombination. Radiative decay rate is measured by ultrafast measurement. Typically on the ~ 10 – 100 ps scale and correlated to diffusion length.

Oscillator Strength: Dimensionless quantity related to absorption or emission of dipole. Excitonic dipole is simply described by Γ_0/ω_0 (where Γ_0 is decay rate and ω_0 exciton resonant frequency).

Exciton Radius: Excitonic Bohr radius which can be described by the hydrogen atom model with effective masses of electron and hole. Wannier–Mott exciton and Frenkel exciton.

[Categorized Types of Exciton]

Trion/Biexciton: Excited states of excitons formed by additional combination with charged carrier or another exciton during photoexcitation.

Bright Exciton/Dark Exciton: Dark exciton is spin-forbidden optical transition, hence it lasts significantly longer than bright exciton (spin-allowed).

Singlet/Triplet: In organic materials, singlet ($S = 0$) and triplet ($S = 1$) excitons are observed. Singlets and triplets are analogous to bright and dark excitons, respectively. Triplet lasts 10^4 times longer than singlet.

low exciton binding energy yet are stable at room temperature. These excitons are observed in 0D quantum dots (QDs), namely, II–IV QDs, perovskite QDs,^{4–6} 1D single-walled carbon nanotubes (SWCNT),^{7–9} and 2D transition metal dichalcogenides (TMDCs),^{10,11} and hybrid organic–inorganic perovskites (HOIP).¹² The excitonic materials covered in this review article are shown schematically in Figure 1B. Briefly, QDs are semiconductor nanocrystals with a tunable band gap by varying the nanocrystal size of the same constituents (*e.g.*, CdS, CsPbBr₃). SWCNTs are carbon allotropes with sp²-hybridized carbon sheets (monolayer graphene ribbons) rolled up into seamless tubes. TMDCs are van der Waals layered materials with a chemical formula of MX₂, where M is the transition metal (Mo, W) and X is a chalcogen (S, Se, Te). HOIPs are lead halide (inorganic PbI₂) quantum wells separated by organic cations (or ligands) stacked to develop a multilayered quantum well material. The number of quantum wells and organic layers in a unit cell defines the order of the HOIP.^{12–16} Different from the exciton property of individual materials, combining two semiconducting materials such as TMDC heterostructure,^{17–19} TMDC/organic structure,^{20–22} TMDC/perovskite QDs,^{23–25} TMDC/HOIP^{26–29} hybrid structure, or even trilayer heterostructures^{30,31} results in the formation of charge-transfer excitons or interlayer excitons which exhibit long-lived states with their dipole moment (out-of-plane) different from their in-plane dipole moment. Recently, careful control over these heterostructures both in-plane and out-of-plane *via* twist and stacking degrees of freedom have allowed the investigation of excitons in

Moiré^{32–36} and hybrid superlattices.^{37,38} Such a variety of excitonic behavior present in the multidimensional semiconductors as discussed above can lead to optoelectronic properties when they are coupled with light by integrating them in an external photonic system. For instance, excitonic materials hybridized with light in an optical mirror enable strong light–matter coupling to form exciton–polaritons, which are half-light, half-matter states. These polariton states are pivotal in the realization of Bose–Einstein condensates for polariton lasing,^{39–41} polariton photochemistry for photocatalytic reactions,^{42,43} entanglement,^{44,45} neuromorphic computing,⁴⁶ and quantum computing,^{47–49} which are some of the emerging topics for studying exciton physics and realizing advanced optoelectronic devices. Without limiting the external medium to a photonic system, these hybrid light–matter states can also be controlled electrically, which expedites their potential for integration into microelectronic systems.^{7,50–52}

This review will focus on understanding the light–matter coupling in ensembles of particles (strong coupling) and a single-particle (weak coupling) system for both Frenkel and Wannier–Mott excitonic materials of varying dimensionalities ranging from 0D QDs and organic molecules to 1D organic chains (polymers) and nanotubes to 2D van der Waals materials such as TMDC monolayers and HOIPs. We present the role of optical nanocavity mode volume in achieving strong coupling when integrated with excitonic materials and provide a comparison to evaluate their merits/limitations of various types of cavities. In addition, we also highlight and discuss recent works which show that strong light–matter interactions can also be achieved by self-hybridization of photons in the absence of any external cavity. The emerging applications and shortcomings in these excitonic systems that form self-hybridized polaritons are discussed. Second, while several prior attempts at reviewing excitonic materials have been made, their photonic properties have only been discussed in the context of far-field spectroscopy.^{53–61} In the past decade, nanostructure engineering in the subwavelength limit has become prevalent due to the widespread availability of high-resolution lithography techniques. This has enabled precise engineering of resonant light–matter interactions at a mesoscopic level by staying agnostic to the electronic structure of the medium and allowing the tailoring of electromagnetic near-field per will such as by designing various types of nanocavities. Therefore, studying excitonic effects in engineered near-field electromagnetic environments and, more importantly, using near-field spectroscopy to spatially map optical signals at deep subwavelength resolution has taken a defining role in shaping this rapidly evolving research area.⁶² In the nanocavity mode, the potential of using near-field spectroscopy to understand exciton dynamics, observing forbidden exciton transitions,⁶³ superradiance in molecular assemblies⁶⁴ and studies emerging in far-field technique are also detailed. Finally, current state-of-the-art device architectures demonstrate electrical control of hybrid excitonic states, and individual excitonic quantum emitters are presented. Figures of merit for materials in both the strong and weak coupling regimes are discussed, and possible research interest from the present understanding of these excitonic materials will be discussed. Recent developments on extending the application of hybrid state in the artificial lattice,^{65–67} optical neuromorphic computing,⁴⁶ exciton–polariton transistors,^{68,69} photochemistry,^{42,43} and quantum emitters in Moiré excitons⁷⁰ are also outlined.

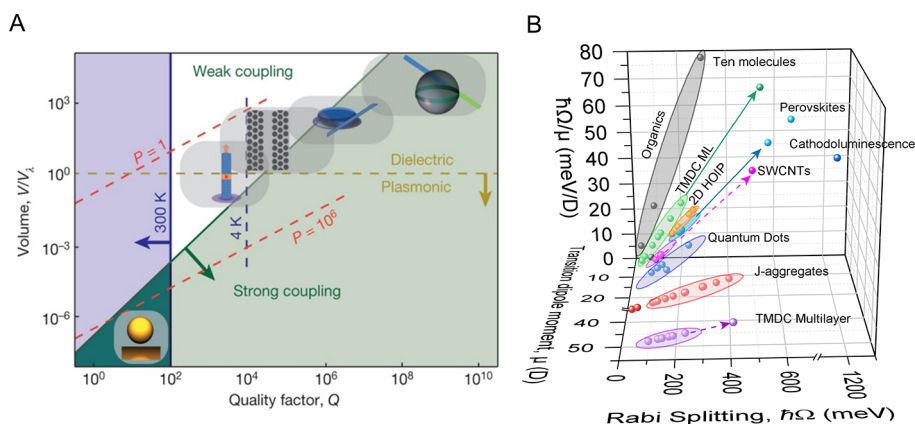


Figure 2. Strong coupling of excitonic materials in a cavity. (A) Comparison of different cavity mode (microcavity, resonator, photonic cavity, and micropillars) shows that the plasmonic cavity can achieve strong coupling at 300 K due to its ultrasmall mode volume despite a low Q factor. (B) Excitonic materials, namely, QDs,^{6,71–75} organic molecules (OD),^{76–78} SWCNTs,^{79–82} perovskites (1D),^{83–86} J-aggregates,^{87–96} 2D HOIP,^{97–100} 2D TMDC monolayers,^{101–109} and 3D TMDC multilayers^{102,103,110–113} studied in cavity mode. Rabi splitting ($\hbar\Omega$) to transition dipole moment (μ) as a figure of merit ($\hbar\Omega/\mu$) is compared for these excitonic materials. Dotted and solid lines show the $\hbar\Omega/\mu$ achieved for excitonic materials where gain medium itself acts as an exciton–plasmon cavity and Fabry–Pérot cavity, respectively. Scatter points outside the shaded ovals represent self-hybridized exciton–polariton systems. Panel A adapted with permission from ref 76. Copyright 2016 Nature Publishing Group.

STRONG COUPLING OF EXCITONIC MATERIALS IN A CAVITY

Optical cavities ranging from Fabry–Pérot (F-P) etalons to photonic crystals and plasmonic cavities typically mediate the coupling of photons with excitons into hybrid modes (Figure 1). Ideally, optical cavities are expected to confine light within the cavity for an infinitely long period of time. However, this is far from the reality due to the nonideal properties of materials and imperfections in cavities. One of the important characteristics of the optical cavity is the quality factor (Q), which is a measure of energy decay within a cavity by absorption or scattering to the environment, and the decay rate is denoted as Γ_{cavity} . Furthermore, the emitter decay (Γ_{emitter}) also influences the strong coupling in an optical cavity. Therefore, the coupling strength (g) must exceed the sum of both cavity decay rate and emitter decay rate ($g > \Gamma_{\text{cavity}} + \Gamma_{\text{emitter}}$). When this condition is taken into account, quantum emitters (InAs QDs¹¹⁴ in photonic crystal cavity) are studied at cryogenic temperatures with large mode volumes (V_m) and high Q factors to achieve strong coupling, as shown in Figure 2A. Although the large microcavity mode volume satisfies the condition for coupling strength to observe strong coupling in these systems, it can be noted that the operating temperature is limited to the cryogenic levels. To achieve room-temperature operation, a high Q factor cavity with reduced mode volume needs to be addressed as it is intrinsically limited by the mode volume ($V_m = \lambda/2n$)³, where λ is the wavelength of light in a vacuum and n is the index of refraction of the medium. This reduced mode volume for dielectric cavities remains a challenge even today, with the advanced solid-state device fabrication largely limited by the refractive indices of the medium. On the emitter contribution, the emitter decay rate is also limited by thermal energy ($k_B T$), which results in low Q factor cavities (less than 100). As the condition for strong coupling scales with $Q/\sqrt{V_m}$, this requirement can be met by drastically reducing the mode volume to less than $10^{-5} V_\lambda$ as shown in Figure 2A.

Photonic crystal nanocavities (PhC) have attracted immense attention in nanophotonics in recent years for their high Q

factors and low losses.^{115–118} Excitonic materials [single quantum dots (InAs) or TMDC monolayers]^{119–123} when placed in the PhC medium at a defect site have shown strong interactions with photons, resulting in lasing at low temperatures (130 K). Strong coupling between excitonic materials and PhC forming exciton–polariton states have also been confirmed with reflectance²⁵ and photoluminescence (PL) spectroscopy.^{124–126} The concept of PhCs when extended from dielectric to plasmonic medium results in strong interactions of plasmons with excitons if the excitonic medium is in close proximity of appropriate integration. Several strategies like placing excitonic materials on a Au mirror or placing Au nanoparticles on the semiconductor or making plasmonic crystal cavities have led to strong light–matter coupling, including observation of plexcitons (plasmon–excitons) at room temperature.^{127–133}

Another strategy for light confinement which has recently gained significant attraction is bound states in the continuum (BICs). BICs are photonic states which remain confined and coexist with a continuous spectrum of radiating waves without leaking out as radiation. These optical properties make BICs confine photons for an infinite lifetime as the Q factor diverges to infinity at the intersection of BIC modes.^{134–136} Although these modes can efficiently trap light in a cavity for a prolonged time, unavoidable nonidealities such as the material absorption, finiteness of the structure, and extrinsic perturbations render them with a limited lifetime, which leads to the term quasi-BIC modes. Similar to other cavity modes, quasi-BIC modes can be tuned across the electromagnetic spectrum to couple with different material/lattice excitations. Using quasi-BICs, several optical properties such as directional light emission,¹³⁷ second-harmonic generation,^{138,139} and light–matter interactions have been studied in excitonic semiconductors. Further, BIC modes have been also used to demonstrate vortex with directional lasing in perovskites^{140,141} and nonlinear polaritons in TMDCs.¹⁴² Although integration of BIC modes for light–matter interaction is in nascent stages, their potential to couple with both electric and magnetic dipoles will make them a next-generation cavity system for tailoring optical emission in excitonic systems.¹⁴³

Plasmonic cavities are emerging nanocavity systems that can form small mode volumes and achieve strong coupling despite their low Q factor. One such technique is nanoparticle-on-mirror (NPoM), where the excitonic material is placed on a gold substrate (mirror) and covered with a nanoparticle. When the plasmonic resonance the wavelength of the nanoparticle overlaps with the absorption of the excitonic material placed in the nanocavity, the condition for strong coupling is satisfied to form hybrid states. This technique is highly simplified in terms of fabrication, whereas deterministic spatial positioning is still a challenge. However, it is a much more reliable and reproducible system than plasmonic bowtie nanoantennas. Nevertheless, the nanosized nature of excitonic materials permits achieving sub-nanometer mode volumes and strong coupling (Figure 2A). Without limiting to far-field studies, the plasmonic nanocavity can be exploited to study exciton dynamics across the lateral dimension, as discussed in the next section.

Excitonic materials from 0D to 1D and up to 3D have different transition dipole moments (μ), which influence the light–matter coupling in a cavity. For reasonable comparison of these materials in a strong coupling regime, the ratio of Rabi splitting ($\hbar\Omega$) to transition dipole moment (μ) is considered as a figure of merit in this review and is shown in Figure 2B. It should be noted that, although the photonic cavity has a very high Q factor, and the plasmonic cavity has small mode volumes, the outcome of Rabi splitting which is dependent on $Q/\sqrt{V_m}$ will be dominated by plasmonic cavity when using $\hbar\Omega/\mu$ as a figure of merit.^{55,144} Nevertheless, in this review, our emphasis in Figure 2B is on the performance metric for different excitonic materials, which is mostly dictated by their transition dipole moment and staying agnostic to cavity type. Second, self-hybridized light–matter coupling without an external cavity is an upcoming field of interest in the community.^{111–113} For instance, a self-hybridized system—perovskite nanowire (1D) with an intrinsic F-P mode shows a Rabi splitting, $\hbar\Omega \sim 564$ meV, in the strong coupling regime.⁸³ In such a case, although the μ of the system is less than 20 D, achieving strong coupling in a microcavity should be noted. This is due to various factors such as number of emitters (N), exciton oscillator strength (f), and V_m affecting g in the F-P cavity:

$$g \propto \sqrt{N \cdot \frac{f}{V_m}}$$

Nanowires with subwavelength dimensions reduce the mode volume drastically and hence enhance the coupling strength. On the other hand, patterned TMDC multilayers with high oscillator strength also show a strong coupling of ~ 400 meV.¹¹¹ For such systems, there is a large spread in oscillator strengths for different dimensionalities of semiconductor materials. Reduced mode volume has not been reported in several published reports of Rabi splitting likely due to the difficulty in estimating accurate field profiles and mode dimensions on such small scales. Here, we will focus on a critical analysis using $\hbar\Omega/\mu$ as a figure of merit to identify the best strategies for achieving strong coupling with photons in this review article.

0D materials such as organic molecules and QDs support Frenkel excitons and Wannier–Mott excitons, respectively.^{145–147} Evidence of strong light–matter coupling has been presented in organic molecules such as methylene blue

positioned in plasmonic nanocavities with their dipole moment along the cavity electric field. Likewise, QDs with random dipole moments that are deterministically placed in plasmonic nanocavities such as bowtie antennas have also shown evidence of strong coupling.^{6,71–74,148} Single-particle probe techniques such as cathodoluminescence were used to probe especially large Rabi splittings (~ 1200 meV) from individual CdZnS/CdS QDs placed in Al nanocavities.⁷⁴ For a detailed account on the strong light–matter coupling in inorganic QDs measured in the far-field, the readers are referred to previously published review articles.^{54,55}

Akin to inorganic QDs, functionalized SWCNTs have also shown evidence of strong light–matter coupling from the reflectance spectroscopy studies.^{79,80,82} SWCNTs placed in plasmonic cavities or etching SWCNT films/bundles into nanoresonators have shown Rabi splitting of ~ 100 meV^{79,80} and ~ 500 meV,⁸² respectively. Also, polariton emission from the hybrid states in SWCNTs was observed at the telecommunication wavelength (1330 nm), which is tunable by varying the SWCNT chirality. Further, the high charge carrier mobility and cavity-free polariton emission render SWCNTs promising for electrically injected polariton lasers.⁷ However, SWCNTs have been plagued with poor PL emission efficiencies in their thin-film form. Overcoming this hurdle is of fundamental importance and presents a research challenge in materials optimization.^{82,146,149}

In contrast to SWCNTs, J-aggregates are visible–NIR emitting self-assembled molecular assemblies where the dipole moment of an individual molecule is coupled, which leads to extended delocalization of excitons throughout the aggregate length (theoretically).^{2,150–152} Such strong dipole–dipole coupling in a J-aggregate results in an increased net transition dipole moment and oscillator strength with narrower absorption/emission line width. Strong light–matter coupling has been observed by optically pumping J-aggregates in a distributed Bragg reflector (DBR) cavity,¹⁵³ metallic dimers/gratings,^{154–156} gold gratings,⁹⁵ Si nanoparticles as Mie scatters,^{157–159} dye-coated plasmonic particles,^{155,160–165} and particle on the metallic substrate^{78,166} as well as electrical injection in J-aggregate/GaN microcavity up to 200 K.¹⁶⁷ In these studies, the J-aggregates with an excitonic absorption peak between 2.25 and 2.48 eV form hybrid states with silver plasmonic particles.^{155,160–165} Although silver supports visible plasmons, the narrow interband transition in aluminum (~ 1.5 eV) allows it to extend the surface plasmon response up to the ultraviolet range and also at a low-cost alternative to silver.¹⁶⁸

Consequently, replacing silver plasmonic particles with aluminum pads and covering them with J-aggregates doubled the Rabi splitting (400 meV). In addition to hybrid state formation, the strong coupling of molecules in a cavity has inverted the singlet/triplet state in organic molecules,¹⁶⁹ delayed fluorescence,¹⁷⁰ and suppressed photo-oxidation.¹⁷¹ These reports point toward evidence that strong coupling of light and excitons can alter the exciton recombination pathways, which in turn influences the optoelectronic device performances as well as photochemistry.

In contrast with 0D and 1D semiconductors discussed above, 2D excitonic TMDCs offer gate (electrostatic) tunability of exciton–trion populations, thereby suppressing trion formation (nonradiative decay channels) and enhancing the quantum yield of the excitonic emission.^{172,173} Such gate tunability also controls the formation of exciton–polaritons versus trion–polaritons for TMDCs placed in a plasmonic

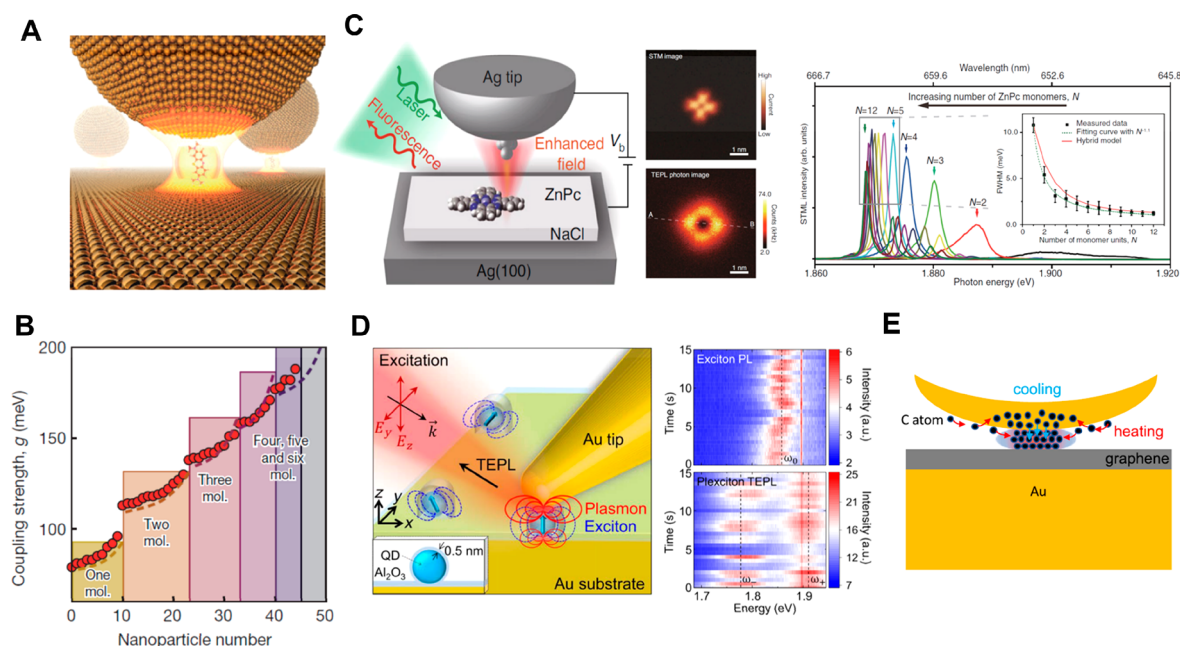


Figure 3. Near-field spectroscopy for 0D and 1D excitonic materials in strong coupling. (A,B) Ultrasmall mode volume in a plasmonic cavity for strong coupling organic molecules showing an increase in coupling with increase in number of molecules. (C) Scanning tunneling microscopy induced luminescence for a zinc phthalocyanine (ZnPc) molecular chain in a plasmonic nanocavity showing imaging at the molecular level. Appearance of a superradiance peak and linewidth narrowing (inset) for different ZnPc chains (right panel). (D) Near-field spectroscopy setup to study QDs in a strong coupling regime (left panel) and photoluminescence spectrum of exciton under weak coupling and plexciton under strong coupling (right panel D). (E) Carbon quantum dot formation and plexciton states under nanocavity mode. Panels A and B adapted with permission from ref 76. Copyright 2016 Nature Publishing Group. Panel C adapted with permission from ref 182 (copyright 2020 Nature Publishing Group) and ref 64 (copyright 2019 American Physical Society). Panel D reprinted/adapted with permission from ref 72. Copyright 2019 The Authors, some rights reserved; exclusive licensee AAAS. Distributed under a CC BY-NC 4.0 license <http://creativecommons.org/licenses/by-nc/4.0/>. Panel E reprinted from ref 190. Copyright 2020 American Chemical Society.

cavity.¹⁰⁴ Moreover, a TMDC monolayer with broken symmetry introduces valley polarization, which is enhanced upon integrating TMDCs with chiral photonic metastructure arrays.¹⁷⁴ Such a property of selective excitation of band valleys in TMDCs by circularly polarized light adds another knob for tuning light–matter interactions when studied in the strong coupling regime.^{174,175} Nevertheless, valley polarization is less pronounced in pristine TMDCs due to strong dephasing. Recent works have shown that the energy degeneracy between σ_+ and σ_- in a TMDC layer can be lifted in the presence of external magnetic fields by virtue of the Zeeman effect and presence of exciton–polaritons under a strong coupling regime.^{176,177} A linear correlation between Zeeman splitting and applied magnetic field (−9 to 9 T) was observed. Finally, the atomic flatness and nanoscale thickness of TMDCs render them particularly suitable for integration into NPoM architecture for strong coupling studies.^{102,103} In fact, the μ of TMDC monolayer (5 D) is lower by 3 orders of magnitude compared to that of the plasmonic Au nanoparticles (1.3×10^4 D). This stark difference motivated the researchers to combine Au nanoparticles with a WS₂ monolayer inside the plasmonic cavity. The collective coupling between plasmon–exciton mode in resonance with the cavity mode led to an enhancement of the Rabi splitting from 75 meV (without plasmonic particles) to 535 meV (with plasmonic particles).¹⁰⁸ Nevertheless, such a hierarchical design will have a small excitonic component in the hybrid state despite reaching ultrastrong coupling. This study can potentially serve as a route to circumvent the intrinsically low oscillator strengths (for example, monolayer TMDCs or organic chromophores) in

excitonic materials that have hindered strong coupling. HOIPs are another class of emerging 2D excitonic media that have also shown strong coupling upon integration in a DBR cavity⁹⁸ or in metasurfaces on SiO₂.¹⁷⁸ Thick HOIP single crystals (without any external cavity) have also shown similar Rabi splitting (170 meV). Cavity-free light–matter hybridization in excitonic media has been discussed in detail.

From the above discussion, we have summarized the strong coupling of excitonic materials from far-field studies. We have discussed several aspects of the strong coupling from exploring plasmonic cavities, intrinsic cavity mode, to probing hybrid light–matter states in various optical cavities. In the next section, we discuss characterization of low-dimensional excitonic materials with near-field spectroscopy techniques wherein we attempt to highlight the physics that is obscured in far-field spectroscopy. In addition, we also discuss cases reporting self-hybridized polaritons from excitonic semiconductors of varying dimensionality.

NEAR-FIELD SPECTROSCOPY AS A PROBE FOR 0D AND 1D SEMICONDUCTORS

Near-field spectroscopy involves an NPoM or metal probe to collect local optical information with a resolution of <20 nm and ultralow cavity mode volumes containing the excitonic medium. Most importantly, scanning near-field techniques provide direct evidence of photophysics at the nanoscale with both high spectral and deep subwavelength spatial resolution in Raman^{179,180} and PL spectra.¹⁸¹ Near-field spectroscopy in plasmonic nanocavities for organic molecules was investigated for methylene blue (0D)⁷⁶ and 1D zinc phthalocyanine

(1D).^{64,182} Encapsulating methylene blue in a cucurbit[7]uril matrix and placing a plasmonic Au particle on top, strong coupling was achieved (Figure 3A).⁷⁶ In this study, the relation between coupling strength (g) and number of molecules (N) in the cavity, $g \propto \sqrt{N}$, was demonstrated. The orientation of the dipole moment of the methylene blue molecule along the electric field of the cavity enhanced the coupling strength up to 200 meV with increasing the number of molecules in the nanocavity (Figure 3B).

Beyond light–matter coupling and achieving hybrid states, near-field nanospectroscopy can also help unravel the fundamentals of dipole moment characteristics, its coupling with the electric field, orientations, and spatial imaging of the molecular structure. Nanoscale imaging of molecules was performed using tip-enhanced Raman spectroscopy (TERS) and nonplasmonic tips (Si) to obviate photoluminescence quenching near the vicinity of the metals. Nevertheless, both TERS and PL imaging have shown a reasonable spatial resolution of molecules of ~ 15 nm¹⁸³ and sub-10 nm,¹⁸⁴ respectively. When the plasmonic nanocavity mode was brought in resonance with the molecular vibrational transitions, spatial resolution of TERS less than 1 nm was achieved by controlling the inelastic scattering.¹⁸⁵ Later, photoluminescence was also probed for imaging electronic excitations at the molecular level with ~ 8 Å spatial resolution.¹⁸² Generally, the transition dipole moment of the molecules is assumed to be a point dipole moment (PDM) in nature. Using tip-enhanced photoluminescence (TEPL), ZnPc molecules can be imaged up to sub-nanometer resolution (~ 8 Å)^{182,186,187} and nanobubbles in TMDCs up to 15 nm^{188,189} with a plasmonic (Ag or Au) tip.

Enhanced PL from the near-field in a plasmonic cavity was used to map ZnPc PL with high sensitivity and resolution. As shown in Figure 3C (right panel), the TEPL signal from ZnPc shows four lobes signifying the four-fold symmetry of the molecule.¹⁸² Direct observation of the two orthogonal dipole moments gives rise to four-fold symmetry and not PDM, which is a fundamentally additional insight revealed by near-field spectroscopy. The dark region in the center was due to the cancellation of the dipole symmetry from the whole molecule. Further, closely packing these ZnPc molecules into 1D nanowire leads to dipole–dipole coupling. Again using TEPL, red-shifted PL peaks with reduced line width as well as signatures of superradiance were observed upon increasing the number of molecules.⁶⁴

Although near-field spectroscopy using TEPL and far-field surface-enhanced Raman spectroscopy have been extensively investigated, unexplored domains such as tip-enhanced Raman spectroscopy can be used to identify nonradiative decay channels, which are dark states in far-field spectroscopy. For instance, using TEPL for nanoimaging and identifying nonradiative spots can also be misinterpreted from photobleaching of organic molecules in ambient conditions. However, the Raman spectrum can show a distinct signature of individual molecules (monomers) and J-aggregates.^{185,191,192} In this way, monomer molecules which act as exciton trap states can be identified to understand the spatial position of the molecules in large size assemblies using TERS.¹⁹³ Further, introducing time-resolved measurements in a near-field technique can be helpful to identify the exciton localization and nonradiative recombination at these exciton traps.

QDs are one of the best model systems to investigate strong and weak coupling due to their high quantum yield and random dipole orientation. Since gap modes create high electric fields along the z -direction, a QD with an excitonic dipole along the z -direction can strongly couple with a plasmonic nanocavity, while another QD with a different dipole orientation can only induce weak coupling. For example, exciton–plasmon polariton coupled states, so-called plexcitons, were reported using a CdSe/ZnS QD on Au with a 0.5 nm dielectric capping layer (Figure 3D).⁷² Multiple QDs showed different types of spectra with respect to degree of coupling between QD excitonic dipoles and gap plasmon modes. In the strong coupling regime, the coupling strength (g) was found to vary from 70 to 163 meV, which reveals plexciton formation due to randomly distributed dipole orientation of each QD. In the case of weak coupling, on the other hand, only photoluminescence enhancement was observed due to the increased radiative rate by the Purcell effect. Similarly, an in-plane plasmonic resonator tip was also investigated with CdSeTe/ZnS QD-embedded poly(methyl methacrylate) medium to observe strong coupling wherein a plasmonic substrate is not necessary.⁶ Likewise, plexcitonic states have also been reported in carbon quantum dots (CQDs) by placing graphene between Au nanoparticles and a Au substrate (Figure 3E). In this case, the graphene inside the nanocavity acts as a template for forming CQDs resembling the graphene lattice. High electric fields in the gap mode trigger carbonization of the preadsorbed hydrocarbons on the Au tip into CQDs. The authors reported an anticrossing (Rabi splitting) of ~ 200 meV in emission, revealing successful formation of plexcitonic states.¹⁹⁰

NEAR-FIELD SPECTROSCOPY STUDY ON 2D MATERIALS

Near-field spectroscopy is particularly helpful for investigating 2D semiconductors as they can have atomically thin thicknesses and can be infinitely large in lateral size. This 2D structure allows it to maintain uniformly confined plasmonic gap modes throughout the entire flake. Within 2D materials, excitonic TMDCs have inspired increased interest in the near-field spectroscopy studies due to their distinct optical properties and emergence of locking in spin and valley degrees of freedom,^{194,195} presence of spatial heterogeneity,^{196–198} strain engineering,¹⁹⁹ and in-plane epitaxial heterolayer structures.^{200–203}

In semiconducting TMDCs, strong spin–orbit coupling leads to spin-dependent energy separation of the conduction and valence bands. For example, Mo-based TMDCs form the lowest conduction band with spin-up states, whereas W-based TMDCs do so with spin-down states. Since the highest valence band consists of spin-up states, the smallest energy optical transition is spin-forbidden for W-based TMDCs, which is termed as dark excitons. Dark excitons have a long decay time in the nanosecond regime because they do not couple easily to emitted light and, hence, are the subject of numerous investigations for quantum information processing.²¹² In contrast, the bright excitons in TMDCs have a lifetime of picoseconds.^{213,214} However, using a plasmonic tip in near-field spectroscopy allows one to turn on radiative decay processes and to control dark and bright excitons in 2D TMDC crystals. Specifically, the plasmonic tip enables direct proof of the existence and radiative control of dark exciton emission in WSe₂ crystals due to its out-of-plane optical

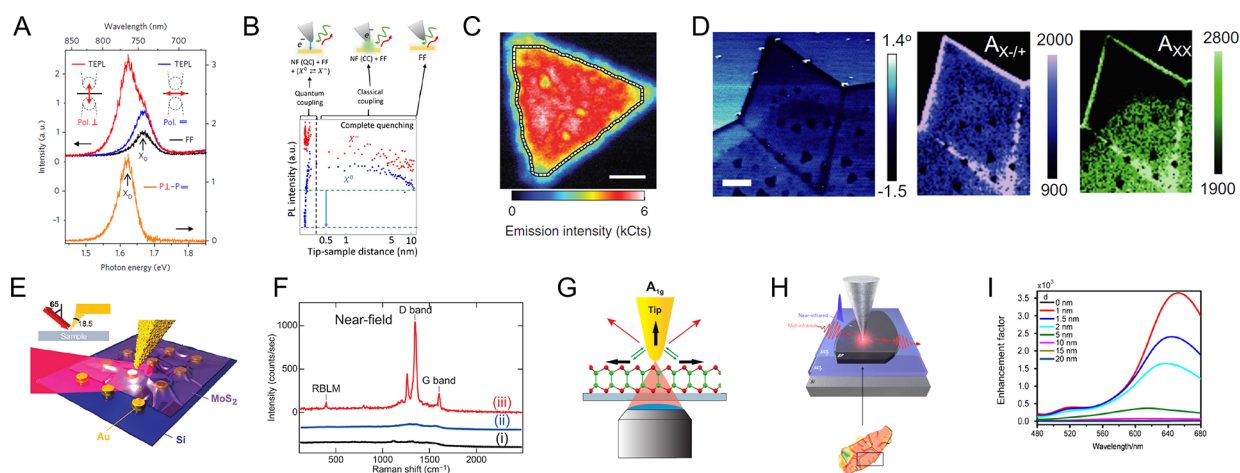


Figure 4. Near-field spectroscopy for 2D materials. (A) Radiative decay of dark exciton on WSe₂ by the Purcell effect. (B) Trion/neutral exciton control on WS₂ using plasmonic picocavity. (C) Nanoscale spatial variation of photoluminescence on MoS₂, exhibiting distinct interior and edge emission. (D) Correlation between Kelvin probe force microscopy phase, trion, and biexcitonic emission mapping of MoSe₂. (E) TERS spectrum of MoS₂ on a Au nanocylinder array. (F) TERS spectrum of graphene nanoribbon (red), polyanthrylene/Au(111) (blue), and clean Au(111) (black). (G) TERS spectrum of GaS to investigate phonon correlation length. (H) Ultrafast switching of interface polariton on 2D black phosphorus. (I) Finite-difference time-domain simulation on TEPL of black phosphorus in a nanocavity. Panel A adapted with permission from ref 204. Copyright 2018 Nature Publishing Group. Panel B reprinted/adapted with permission from ref 205. Copyright 2019 The Authors, some rights reserved; exclusive licensee AAAS. Distributed under a CC BY-NC 4.0 license <http://creativecommons.org/licenses/by-nc/4.0/>. Panel C adapted with permission under a Creative Commons CC BY License from ref 206. Copyright 2015 Springer Nature. Panel D adapted with permission under a Creative Commons CC BY License from ref 207. Copyright 2020 Springer Nature. Panel E reproduced with permission from ref 196. Copyright 2018 The Royal Society of Chemistry. Panel F adapted from ref 208. Copyright 2014 American Chemical Society. Panel G reprinted from ref 209. Copyright 2019 American Chemical Society. Panel H adapted with permission from ref 210. Copyright 2017 Nature Publishing Group. Panel I reprinted with permission from ref 211. Copyright 2019 Wiley Publishing Group.

transition (Figure 4A).²⁰⁴ The radiative decay of a dark exciton requires a spin-flipping process typically induced by a large magnetic field and records 10^2 – 10^3 times smaller intensity compared to that of the bright exciton.²¹⁵ However, in the vicinity of the tip, the dark exciton is more emissive due to the increased excitation rate of the dark exciton by the out-of-plane field enhancement as well as enhanced radiative decay rate due to the Purcell effect. Further, the tip controls dark exciton emission by changing the plasmonic gap between 1 and 5 nm. As the tip–sample distance decreases, high emission was achieved due to the enhanced quality factor of the gap mode. Further, by controlling the cavity at the picoscale, it was able to control charged exciton (trion) formation (Figure 4B). Under this small separation between the tip and substrate that forms a picocavity for plasmon confinement, the tip can also tunnel charge into the WS₂, leading to trion formation.²⁰⁵ These results verify that a scanning plasmonic tip can form an active cavity with small mode volumes that enables control over various types of excitonic emissions from 2D TMDCs.

Nanoscale spatial resolution in near-field spectroscopy is further important in resolving crystal defects and interface heterogeneities in 2D TMDCs. The technique has been used to probe features such as defects, grain boundaries,^{197,216} disordered edges,²¹⁶ oxidized surfaces by aging effects,²¹⁷ electronic contact quality,²¹⁸ and strain.^{188,189,196,219} Although these structural heterogeneities are vital for understanding 2D semiconductors and optimizing optoelectronic devices comprising them, they cannot be resolved by diffraction-limited far-field spectroscopy tools due to their physical sizes and interaction with visible frequency photons. Hence, near-field spectroscopy plays an essential role in investigating these nanoscale features. The earliest reports of near-field spectroscopy

use aperture-type campanile tips, which can resolve by ~ 60 nm features revealing differences between disordered edge emitting regions and interior regions in chemical vapor deposition (CVD)-grown monolayer MoS₂ (Figure 4C).²¹⁶ Specifically, ~ 300 nm of the peripheral edge region was discovered with disordered, low intensity, broad energy emission, whereas the interior region showed higher energy and intensity emission due to reduced trion formation. Inspired by this initial study, similar studies have been conducted with an apertureless plasmonic tip. Since the apertureless tips utilize the Purcell effect to confine high electric fields in nanocavities, tip radii determine the resolution which is typically ~ 20 nm. Using an apertureless tip, similar spatial variation in photoluminescence in CVD-grown monolayer MoS₂ crystals was attributed to point defects, nanoscale terraces, and edge emission.²²⁰ For example, localized exciton emission from nanoscale bubbles of monolayer WSe₂ with diameters of ~ 50 nm was observed.¹⁸⁹ Because bubbles induce a slightly smaller optical band gap surrounded by continuous excitonic medium of TMDCs, an exciton funneling effect arises, leading to brighter emission at the bubble. Strain, which is an important parameter of the bubbles, was quantitatively estimated by solving strain tensors using topography and Poisson's ratio, which coincides with strain evaluation from TERS.¹⁸⁹ Due to the operation at room temperature, the strained nanobubbles in 2D TMDCs are particularly interesting from the perspective of single-photon emission (SPE), as discussed. Another example of 2D excitonic semiconductors where near-field spectroscopy is particularly useful is on artificially grown lateral heterojunctions and superlattices of TMDCs like MoSe₂/WSe₂.^{201,221,222} Owing to the similarity in lattice constants and crystal structures,

combined with their atomically thin structure, it is feasible to make in-plane stitched, lateral heterointerfaces between two different 2D semiconductors of varying band gaps. The junction comprises a depletion region as well as a strain, which are of deep subwavelength dimensions. In addition, the as-grown lateral MoSe₂/WSe₂ heterostructures exhibit sub-wavelength periodic band gap changes of MoSe₂ and WSe₂. Using TEPL profile, the junction width was estimated as ~150 nm, which is on the subwavelength scale.^{201,217,221} TEPL analysis has also provided evidence of directional hot-electron injection from WSe₂ to MoSe₂ by potential gradient at the junction, which results in enhancement of the luminescence in MoSe₂ regions. Another big advantage of using a plasmonic metal tip is a correlation with nondestructive electrical measurements such as potential, capacitance, and electric and photoconductive measurement. Correlation between TEPL, TERS, potential, and second derivative capacitance maps on MoSe₂ revealed nanoscale inhomogeneities.¹⁹⁷ The capacitance change under illumination/dark environment also proved that grain boundaries can be metallic twin boundaries.²²³ Further, optical properties of the oxidized MoSe₂ monolayer have also been verified by correlating Kelvin probe force microscopy phase to photoluminescence maps (Figure 4D).²⁰⁷ Definitive proof of oxidation-induced dielectric environment change and doping, which results in enhanced biexciton formation, was obtained from this correlation between optical and electronic scanning probe measurements. Scanning probe techniques coupled with light illumination have also facilitated probing buried interfaces of excitonic materials such as TMDC–metal contacts where the TMDCs are directly exfoliated on flat metals. When combined with local photocurrent measurements and resonant Raman spectroscopy, recent studies have shown spatial inhomogeneity in the WSe₂–metal contact interface.²²⁴ The probing of TMDC–metal buried contact interfaces can be further extended to include directly evaporated contacts *via* a Au-assisted transfer technique.²²⁵ This transfer technique flips the buried TMDC–metal interface, exposing it to the sample surface. Further, the contact interfaces created by direct metal evaporation are clearly distinguished from typical TMDC–metal interface fabricated by direct exfoliation process *via* scanning conductance, potential, photoluminescence, and Raman probes. This has not only allowed comparison of TMDC–metal contacts that are formed *via* different evaporation/exfoliation schemes but also allowed investigation of different TMDC/metal combinations in determining the lowest resistance contact making schemes.²¹⁸

Near-field Raman spectroscopy is also an important spectroscopy mode for studying 2D TMDCs since it provides complementary information as compared to PL, such as nanoscale inhomogeneities in charge doping, stoichiometry, and strain. For instance, TERS measured from polycrystalline MoSe₂ transferred on a Au substrate showed nanoscale variance in A₁' and E' mode intensity at the basal plane and grain boundaries.¹⁹⁷ This variance was attributed to local doping change as a result of oxidation of the selenide. Since high signal-to-noise ratios are challenging to obtain in TERS, various types of surface preparation techniques have been developed to enhance TERS signals in practice. For instance, certain TERS studies of MoS₂ on Au nanoarrays have been used as cylindrical (Figure 4E) or triangular structures/shapes of Au.^{196,219} An 8-fold enhancement of TERS signal at the edge of the Au clusters was observed. Under the nonresonant

TERS conditions, a large intensity enhancement of the out-of-plane A_{1g} mode was found due to strong E-field in the nanocavity, whereas the in-plane E_{2g} mode was suppressed.

Graphene is the prototypical 2D material which exhibits semimetallic properties with no band gap. However, confining graphene to <10 nm dimensions opens up a band gap and enables it to support excitons, akin to carbon nanotubes.²²⁶ Accordingly, near-field optical imaging has revealed not only defects and doping in graphene^{227,228} but also properties in confined graphene such as edge-induced localized electronic states.²²⁹ Graphene nanoribbons (GNRs) have been extensively studied due to their size-dependent band gaps.^{230–232} However, due to their small sizes, the edge states dominate transport in GNRs, which motivated the investigation to study edge states of the nanoribbons *via* near-field spectroscopy (Figure 4F).²⁰⁸ The edge states were investigated by spatially mapping the D peak intensity, which mainly arises from the armchair-type edges.²³³ The phase-breaking length, which represents the average distance of excited electron–hole pair diffusion, was observed to be 4.2 nm, which was a far smaller value compared to the estimate from far-field measurement (~50 nm). This discrepancy is once again a result of the large beam size of the Gaussian beam of the far-field measurement, which is not able to resolve ~4 nm width and therefore highlights the power and utility of near-field scanning optical nanoprobes.

Aside from TMDCs of Mo, W, and GNRs, groups III–VI layered semiconductors MX (M = Ga and In; X = S, Se and Te) are also important members of the layered materials family, showing relatively large optical band gaps for monolayers (~2.5–3.0 eV).^{209,234} These large optical band gaps have attracted their use in applications such as blue light-emitting diodes.²³⁵ However, low emission efficiencies and weak Raman signals in monolayers present challenges in investigating them *via* far-field spectroscopic techniques.^{235,236} Therefore, tip-enhanced near-field spectroscopy or coupling with plasmonic arrays can offer a viable strategy toward enhancing Raman or PL signals from these materials.^{209,237} In this regard, a recent study reported estimating phonon correlation lengths (*L_c*) of 50–60 nm in few-layered GaS with a plasmonic Au tip (Figure 4G).²⁰⁹ *L_c* was measured by collecting the Raman spectrum as a function of tip–sample distance. Although two A_{1g} modes have identical symmetry, high-order phonon–phonon coupling in A_{21g} leads to a decrease in TERS enhancement. This result is striking difference from graphene since phonon–phonon coupling and mode symmetry influence the Raman scattering in GaS. Near-field studies on III–VI layered semiconductors are currently in early stages. Further, the exciton binding energies in this class of 2D semiconductors are small.²³⁸ Hence, it is difficult to observe strong exciton–photon coupling phenomena at room temperature in near-field conditions. Nonetheless, they are valuable from an application perspective as they maintain a direct band gap even in the few layers to the bulk thickness limit. Therefore, nano-resolution TEPL, defect, strain analysis will be helpful in understanding their optical and electrical properties and develop future applications out of them.

While III–VI layered semiconductors are on the wider gap side, 2D black phosphorus (BP) is another promising layered van der Waals semiconductor for optoelectronics, albeit on the narrow gap side. Its band gap is tunable from 0.3 eV in the bulk to 1.5 eV for monolayers, which covers visible to infrared

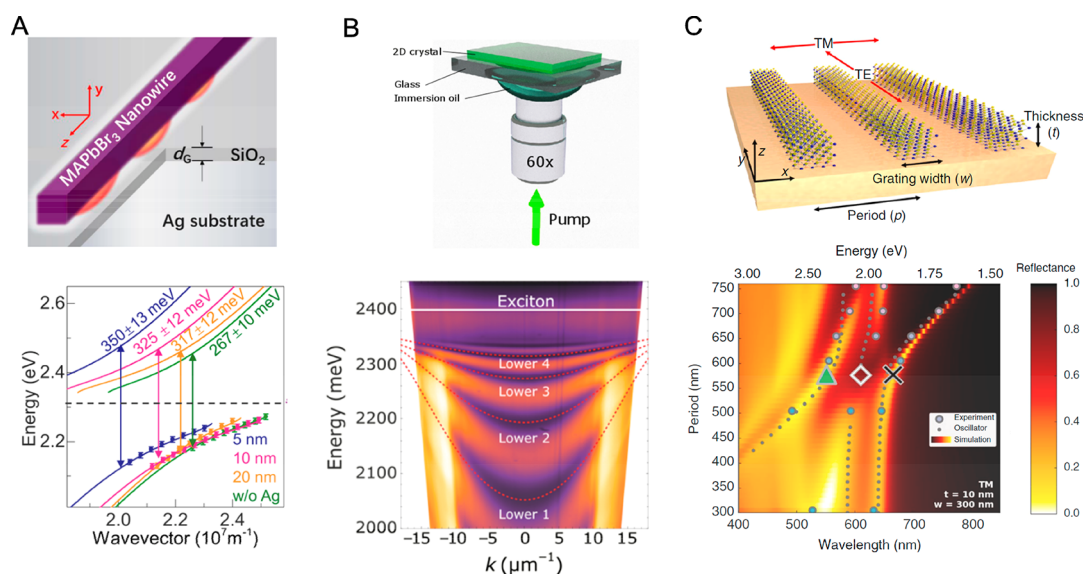


Figure 5. Far-field spectroscopy studies for excitonic materials in open cavity systems. (A) Perovskite nanowire in an open plasmonic cavity. Rabi splitting was observed from strong coupling by decreasing the distance between the perovskite nanowire and silver cavity. (B) 2D HOIP single crystal on glass substrate showing Rabi splitting. (C) Multilayer WS₂ grating structure (thickness, $t = 10$ nm, width, $w = 300$ nm) with varying the period (p) on a gold substrate showing strong coupling with increasing WS₂ thickness. Panel A reprinted from ref 83. Copyright 2018 American Chemical Society. Panel B reprinted/adapted with permission from ref 98. Copyright 2019 The Authors, some rights reserved; exclusive licensee AAAS. Distributed under a CC BY-NC 4.0 license <http://creativecommons.org/licenses/by-nc/4.0/>. Panel C reprinted with permission under a Creative Commons CC BY License from ref 111. Copyright 2020 Springer Nature.

ranges,^{239,240} and it has been shown to have applications in terahertz photodetectors²⁴¹ as well as shown to have a plasmonic response.²⁴² Interface polaritons in multilayer BP with ultrafast photoswitching of ~ 50 fs was reported by using scattering-type near-field with infrared laser (Figure 4H).²¹⁰ While experimental studies in near-field spectroscopy in BP are not available yet, a recent calculation predicted enhancement factors up to 10^8 and 10^3 for Raman and PL, respectively, in BP by using a 3D finite-difference time-domain method (Figure 4I).²¹¹ Still, investigation on BP is in the early stages. Monolayers present an opportunity to study exciton–photon coupling in the near-field particularly given the in-plane anisotropy of the excitons,²⁴³ but the fast oxidation rates under ambient conditions require encapsulation, which presents significant experimental challenges.

In the above section, we have reviewed and discussed how near-field spectroscopy studies help in understanding the photophysics from multidimensional semiconductor materials particularly with regards to exciton–photon interactions and coupling. Additional insights into exciton dynamics at the nanoscale level unraveled by the high spatial resolution of TEPL and TERS highlight the importance of near-field studies.

FAR-FIELD SPECTROSCOPY IN OPEN CAVITY SYSTEMS FROM BULK SEMICONDUCTORS

While the tip serves as a cavity or resonator for all the near-field investigations discussed above, an external cavity is not necessarily required to exploit light–matter coupling. Recent advances have shown that the gain medium can itself act as an open cavity and negates the need for an external cavity. Extraordinary optical constants inherent in materials such as colloidal QDs, perovskites, and TMDCs can sustain F-P (open) cavity mode which leads to strong light–matter coupling without any external cavity in subwavelength thicknesses of these media. For instance, depositing low

concentration 0D QDs on a Si/SiO₂ substrate shows polariton emission, as confirmed from the PL splitting.²⁴⁴ These polariton states are formed from the BIC modes at the interface between Si and SiO₂, supporting a low-refractive/high-refractive index. Coupling BIC modes in a photonic crystal cavity with TMDCs has been shown to enhance polariton nonlinearities.^{142,245} Perovskite nanowires on a Ag substrate have shown strong Rabi splitting due to a combination of reduced mode volume and high oscillator strength (Figure 5A, bottom panel).

Without limiting them to 1D nanowires, HOIP in 2D single crystals also show Rabi splitting in an open cavity (Figure 5B). Thick HOIP crystals can sustain the polariton and interact strongly with the exciton ($3 \pm 0.5 \mu\text{eV} \mu\text{m}^2$) at room temperature. Further, the presence of a Rashba effect in HOIP introduces spin-dependent polaritons, which are beneficial for applications in spintronics and electrically controlled polariton formation devices.^{246–249}

Multilayer TMDCs are another exciting group of materials to realize cavity-free strong coupling of excitons and photons. Several recent studies have shown that metastructures such as disks,¹¹² gratings,¹¹¹ or bottom-up synthesized nanotubes²⁵⁰ can help support coupled exciton–photon states. These metastructures form a F-P cavity mode and interact with excitons to create hybrid states.^{111,112,250} Shaping thick WS₂ layers into disks forms anapole modes, whereas free-standing thick films support F-P modes that couple to excitons.^{112,251} When placed on a reflecting metals such as Au, Ag, or Al, the thickness of the multilayer WS₂ flakes that couple to the F-P (cavity) mode can be substantially reduced (<20 nm), which concurrently induces near-unity absorption in these ultrathin semiconductor samples. Further, etching gratings in these multilayer WS₂ crystals lead to the excitation of a dielectric grating mode in addition to a surface-plasmon-guided mode. Excitons strongly interact with cavity mode and splits into UPB

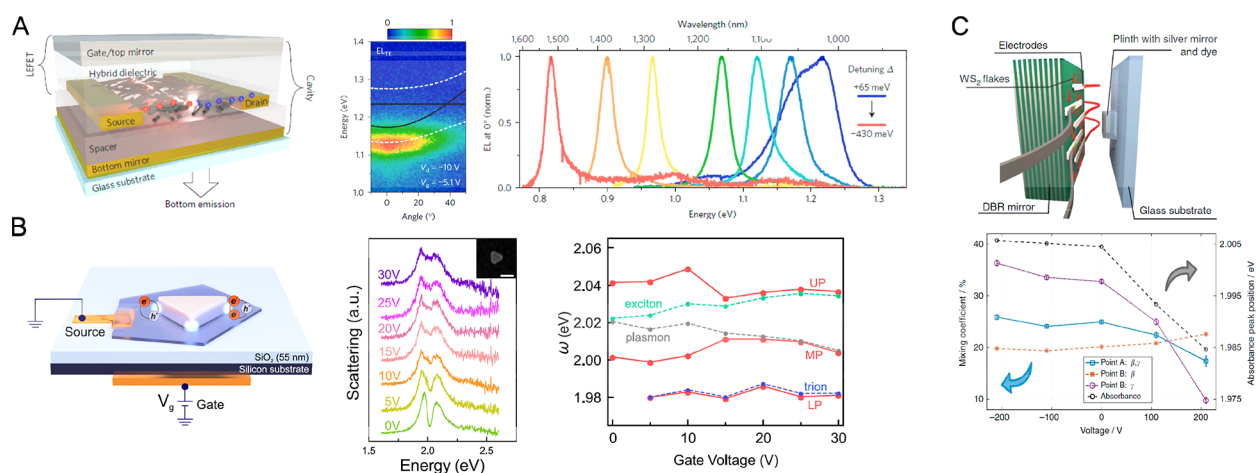


Figure 6. Electrical control of strong light-matter coupling. (A) Device structure of bottom contact/top gate light-emitting field-effect transistor (LEFET) using (6,5) SWCNTs. Angular reflectivity of the channel area in the LEFET showing Rabi splitting in electroluminescence (middle panel) and angle-resolved electroluminescence with detuning the cavity (right panel). (B) Hybrid plasmon-exciton in monolayer WS_2 under electrostatic gating. Dark-field scattering showing hybrid state tunability with gate voltage (middle panel). Eigenenergies of the coupled plasmon-exciton-trion Hamiltonian vs gate voltage at 300 K and atmospheric pressure. (C) Electrically tuned hybrid WS_2 -organic system in a dielectric microcavity showing polariton composition (fraction of Frenkel and Wannier-Mott excitons) along with absorbance at different voltage. Panel A adapted with permission from ref 7. Copyright 2017 Nature Publishing Group. Panel B reprinted from ref 257. Copyright 2020 American Chemical Society. Panel C reprinted with permission under a Creative Commons CC BY License from ref 51. Copyright 2017 Springer Nature.

and LPB. Subsequently, the dielectric grating mode interacts with UPB and LPB to form a three-oscillator coupling with Rabi splitting of ~ 410 meV.¹¹¹ By increasing the grating period, the hybrid modes are tuned where the grating mode emerges into the nondispersive UPB, whereas the LPB switches from a nondispersive to a plasmonic mode (Figure 5C, bottom panel). Although the multilayer TMDC flakes can show strong Rabi splitting,¹¹¹ the indirect nature of the band gap in multilayers limits their characterization with emission spectroscopy. Therefore, emission spectroscopy of self-hybridized polaritons in 2D systems is relatively unexplored and demands the need for van der Waals materials that can maintain direct band gap and high emissivity in the bulk^{252,253} or to explore emissive multilayer TMDCs (for example, ReS_2).²²⁴

In summary, open cavity and intrinsic cavity systems are very appealing as they allow design and realization of monolithic exciton-photon optical elements. Further, their design simplicity will also support electrode integration for electrostatic tuning or electrically injected emission. Another important attribute of the open-cavity systems is their accessibility to other media. For example, for driving polariton-induced chemistry or catalyzing a reaction, an open cavity system allows easy access for the reactants and has the potential for scale up to larger volumes and length scales for photochemistry and photocatalysis. This is potentially difficult to achieve in a closed nanocavity system. Given that several excitonic materials such as SWCNTs, perovskites, and multilayer TMDCs have shown feasibility in supporting polariton modes in open cavities, this area of research is promising. However, there are several challenges with the concerned excitonic materials such as their intrinsic low quantum yield, stability under ambient conditions, *etc.* This will limit their applicability to very specific classes of devices or other applications. Further, material quality and uniformity as a whole needs to improve, and thus there is plenty more work to be done in that regard. Self-hybridized J-aggregates have shown

Rabi splitting (~ 18 – 40 meV) smaller than that of their counterparts in an external cavity, owing to the low refractive index of the medium.²⁵⁴ Additional strategies such as nanopatterning to form resonators are required to enhance coupling of photons with excitons in such self-hybridized systems.²⁵⁵ Nevertheless, the quantum yield of the J-aggregates requires further enhancement to achieve polariton lasing at room temperature.¹⁹³ Finally, understanding of electrical contacts to these materials and developing strategies for stable and reliable contact making will be important moving forward as it will facilitate seamless integration into active devices.

ELECTRICAL CONTROL OF EMISSION FROM HYBRID EXCITON-PHOTON STATES

Hybrid states formed upon optical pumping of excitonic materials in an external cavity can be generated by external carrier injection in the semiconductors. One of such architectures is the light-emitting field effect transistor (LEFET). The possibility of long-range charge transport in the semiconductors and emission in the center of the channel makes LEFETs more efficient emitters than diode architectures, where excitons are quenched at the metal electrode. A LEFET was fabricated by dispersing a (6,5) SWCNT in a polymer matrix sandwiched between DBR mirrors with the bottom electrode and top gate (Figure 6A). Polariton emission emerging from LEFET with Rabi splitting of ~ 125 meV can be tuned from 1060 to 1530 nm by detuning the cavity (middle panel, Figure 6A). While polariton emission was evident, polariton lasing was not observed even at high injection currents.^{7,256}

Electrostatic gating is a widely used approach for controlling device performance in microelectronics. High photoluminescence quantum yield was achieved for a TMDC monolayer in cavity-free mode by varying the exciton and trion proportion using electrostatic gating.¹⁷³ In a similar architecture, adding a plasmonic nanoparticle on the TMDC monolayer (WS_2) can introduce exciton-plasmon hybrid states (Figure

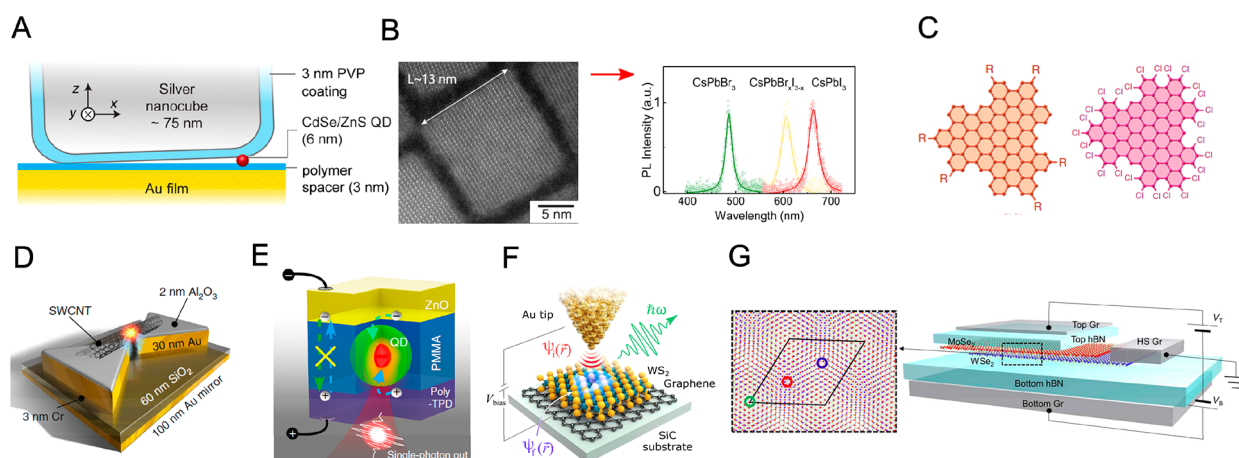


Figure 7. Excitonic single photon emitters. (A) II–IV (CdSe/ZnS QD) placed in plasmonic nanocavity. (B) Perovskite (PVSK) QDs observed under transmission electron microscope and tunable PL by varying the halide ion composition (right panel). (C) Functionalized graphene QDs. (D) SWCNT suspended across a bowtie antenna. Schematic of electrically controlled SPE from (E) CdSe/CdS QDs, (F) tunneling electron-induced emission from WS₂ defects, and (G) Moiré heterobilayer (MoSe₂/WS₂) with dual gates. Panel A adapted from ref 259. Copyright 2016 American Chemical Society. Panel B reprinted from ref 260 (copyright 2015 American Chemical Society) and ref 261 (copyright 2019 The American Association for the Advancement of Science (AAAS)). Panel C reprinted with permission under a Creative Commons CC BY License from ref 262. Copyright 2018 Springer Nature. Panel D reprinted with permission under a Creative Commons CC BY License from ref 8. Copyright 2017 Springer Nature. Panel E reprinted with permission under a Creative Commons CC BY License from ref 263. Copyright 2017 Springer Nature. Panel F reprinted/adapted with permission from ref 264. Copyright 2020 The Authors, some rights reserved; exclusive licensee AAAS. Distributed under a CC BY-NC 4.0 license <http://creativecommons.org/licenses/by-nc/4.0/>. Panel G adapted with permission from ref 265. Copyright 2020 American Association for the Advancement of Science (AAAS).

6B).^{104,257,258} Tuning these hybrid states with electrostatic gating at different gate bias was demonstrated from reflectance spectroscopy. At 77 K, plasmon–exciton becomes dominant ($V = -12$ V). However, excitons remain dominant at room temperature ($V = 30$ V) (Figure 6B). On the other hand, a multilayer TMDC with plasmonic particles shows a UPB and dispersionless collective mode with a tunable polariton band gap.^{49,229}

Forming a hybrid exciton state by mixing Frenkel and Wannier–Mott excitons is an exciting possibility using strong light–matter coupling and making heterostructures of two or more excitonic materials. Such a heterostructure was realized using a combination of organic (J-aggregates) and inorganic (WS₂) excitonic material. Exciton absorption from WS₂ and J-aggregates is spectrally separated by 100 meV. However, when both are strongly coupled in the DBR microcavity, it leads to observation of three polariton branches—a UPB, a LPB, and a middle polariton branch (MPB) which contains Frenkel, Wannier, and Frenkel–Wannier mixed excitons, respectively. However, when the Stark effect is used to tune the lines, only the LPB (WS₂) was tunable, whereas the UPB (organic) absorption remained unaltered. By detuning the cavity length, the polariton’s composition (Hopfield coefficients) in the MPB was tuned to be either Frenkel-rich (β) or Wannier-rich excitons (γ), as shown in Figure 6C, bottom panel. Rabi splitting was also shown to be tunable in WS₂ from 57 to 46 meV by applying an external voltage. Hence, the mixed exciton composition in MPB was modulated only by WS₂.⁵¹ Nonetheless, the possibility to create and tune mixed excitons can be used for exploration in other material combinations as well as device applications.

In the above section, we discussed electrical control of hybrid exciton–photon states ranging from Frenkel excitons in SWCNTs to Wannier–Mott excitons in TMDCs to Frenkel–Wannier–Mott hybrid excitons in TMDC/organic heterostructures. TMDCs have thus far shown a pronounced effect in

hybrid state modulation by electrical control. This is mainly because of their 2D nature with high degree of crystallinity. In the future, research efforts need to be focused not only on how to enhance coupling and electrical tuning but also on how to enhance radiative emission of the coupled states. Thus far, all discussed systems pertain to ensembles of excitons and photons that couple to each other. In the following section, we will focus our discussion on the exciton–cavity interaction in the single-particle or quantum limit achieved by confining and exciting only a few particles in the semiconductor medium.

SINGLE-PHOTON EMISSION FROM LOW-DIMENSIONAL SEMICONDUCTORS

When the two-level energy system of the semiconductor materials integrated into a cavity remains unaltered due to the electromagnetic field inside the cavity while the local density of optical states increases, the radiative recombination of the exciton increases significantly. This spontaneous emission with reduced exciton lifetime is called the Purcell effect. The Purcell effect, while valuable across all domains and applications in optics and optoelectronics, is particularly critical and valuable in the quantum limit since it helps enhance the rate of emission of single-photon emissions from quantized emitters. This is particularly critical as bright and pure single-photon sources are paramount for any photon-based quantum information application. Here, we focus on advancements in II–IV QDs,²⁵⁹ perovskite QDs (PVSK QDs),^{260,261,266,267} graphene QDs (GQDs),²⁶² SWCNTs,^{8,9,268} and TMDCs^{269–281} as the quantum emitters and their integration with optical cavities to produce exciton–photon interactions. Specifically, we review their properties and understand how they compare with III–V QDs,^{282–284} which are the workhorse of SPE. SPE performance such as second-order autocorrelation ($g^2(0)$), lifetime, and the operating temperature is pivotal to realize device integration. Low $g^2(0)$ and achieving room-temperature

operation are the most sought properties in SPE candidates for device integration. In terms of emitter lifetime, two different applications are typically sought. Short exciton lifetime can be used for a high-bit rate single-photon source, whereas longer lifetime emitters are sought for use in quantum memory devices.

SPE from QDs is studied by either placing the emitters in a plasmonic cavity, diluting them (PVSK QDs) to a single-particle limit, functionalizing in the case of GQDs, and placing SWCNTs on bowtie antennas, as shown in Figure 7A–D.^{259–262,266,267} In TMDCs, excitons are localized by inducing strain on the TMDC monolayer at the plasmonic particle edges, or excitons trapped in Moiré lattice act as the SPE source. Further, for electrical control of SPE from QDs, defects in TMDC and Moiré excitons are realized, as shown in Figure 7E–G. We will first discuss the SPE from the plasmonic nanocavities and later present electrically controlled SPE, swiftly overcoming optical excitation challenges while realizing an electronic device.

II–IV QDs of CdSe/ZnS were the initial study demonstrating the Purcell effect to realize SPE in a plasmonic nanocavity (Figure 7A). A 540-fold decrease in exciton lifetime and 1900-fold increase in the total emission intensity of the SPE from these II–IV QDs were reported.²⁵⁹ PVSK QDs are another family of 0D materials where the halide ion composition can be easily tuned to vary the band gap and emission emanating from them (Figure 7B). Nevertheless, the fundamentals of exciton dynamics in PVSK QDs are much more complicated than those of II–IV semiconductors. Rashba effect and exchange interaction-induced exciton splitting into singlet and triplet states complicates the emission from PVSK QDs.²⁶¹ Due to these factors, PVSK QDs show a bright triplet state as the lower-lying state with accelerated radiative recombination (~ 100 ps),²⁸⁵ whereas CdSe QDs are optically dark with a long lifetime (microseconds) at low temperatures. By drop-casting weakly dispersed PVSK QDs on a glass substrate and using single photon spectroscopy, SPE was identified at both 6 K²⁶⁶ and room temperature.²⁶⁰ Single-photon spectroscopy studies at 6 K showed a drastic decrease in the photoluminescence linewidth (~ 1 meV) with reduced blinking. Peculiar to QDs, additional decay channels such as non-radiative Auger recombination and radiative multiphoton recombination remain unavoidable. Although at low fluence (10 μ W) the SPE is devoid of Auger recombination with a short lifetime (180–250 ps), at high fluence, multiphoton emission becomes dominant.²⁶⁶ At room temperature, non-radiative Auger recombination was exploited to selectively suppress multiphoton emission to achieve high-purity single photons under continuous wave or pulsed excitation.²⁶⁰ This is an interesting approach to quell additional decay channels and achieve pure SPE. SPE from QDs can also be extended to purely organic systems. Recently, chlorine-functionalized GQDs have shown SPE at room temperature with high photon purity (0.1) and long lifetimes (5 ns).²⁶² Given the flexibility in designing molecules and adding functionalities to GQDs, SPE with wavelength tunability in the visible region will be of great interest to explore in this graphitic carbon system.

In addition to 0D graphitic QDs, SPE is also possible from 1D nanotubes. Pristine 1D SWCNTs, when placed on plasmonic bowtie antennas, have shown SPE along with a quantum yield enhancement by 62% due to the Purcell effect (Figure 7D).⁸ Subsequently, functional groups covalently

bonded to SWCNTs have led to sp^3 defect formation, which was shown to localize the excitons in an open cavity.^{9,286,287} High photon purity (~ 0.01) with emission in the near-infrared region (1200–1500 nm) at room temperature from SWCNTs was demonstrated, which are attractive properties for applications in the telecommunications domain.⁹

SPE from 2D TMDCs is observed only in monolayer devices due to their direct band gap nature. So far, SPE in TMDCs is realized in several ways such as localized excitons,^{272–275} QD formation from TMDCs,^{276,288} electrical stimulus,^{278,279,289} and deterministic strain by nanopillar in a plasmonic nanocavity.^{271,277} Out of all of these approaches, strain-induced QD formation in TMDCs has been extensively studied for SPE. Using near-field spectroscopy, nanobubbles in WS_2 /hBN have shown localized excitons with red-shifted emission (150 meV) compared to the primary excitons.¹⁸⁹ From previous studies, red-shifted emission was generally ascribed to SPE from localized excitons in WSe_2 ^{272–275} and WSe_2 in plasmonic nanocavities.^{270,280,290} Interestingly, the localized exciton dipole direction is out-of-plane compared to primary excitons (in-plane).²⁹¹ Similar observations of out-of-plane exciton dipoles can explain the origin of SPE from strained-induced TMDC QDs which needs further exploration and verification.

Now we will discuss electrical control of SPE from excitonic materials. Electrically driven SPE was studied by sandwiching CdSe/CdS (without any cavity) between hole (poly-TPD) and electron (ZnO) conducting layers for electrical contact (Figure 7E).²⁶³ Electrical excitation showed higher-purity single photons (0.05) than optical excitation (0.08). Previously, we discussed that multiphoton emission becomes dominant at high fluence. Electrical injection suppresses this additional radiative biexciton formation, which is an indirect process. Fast recombination of trions inhibits biexciton (addition of hole/electron to trion) formation, thereby making single exciton recombination the dominant process for SPE. Also, increasing the thickness of the QD shell suppresses the charging effect during electrical excitation.²⁹² Collectively, room-temperature electrical excitation of QDs in a device structure with high-purity single photons is a significant leap for QD-based SPE device technology.

In contrast to SPE achieved by direct electrical excitation of QDs as discussed above, TMDC monolayers require further manipulation to localize excitons by introducing defects or strain. Several strategies toward localizing excitons such as the use of 2D/3D junctions of TMDCs/piezoelectric actuator to control strain,²⁶⁹ controllably introducing atomic defects,^{264,293} and forming interlayer excitons in Moiré potentials²⁶⁵ have been demonstrated to achieve SPE in TMDCs as detailed below. Piezoelectric actuators with WSe_2 monolayer on barium titanate were used to modulate strain by applying an external voltage. By tuning voltage, the strain can be modulated from compressive to tensile up to 0.15% strain in the TMDC. Alternatively, electrons are injected into the defect energy levels in TMDCs using scanning tunneling microscopy (Figure 7F). These defects can be either sulfur vacancies (V_{S_2}) or chromium doped in W site (Cr_W). By applying electric bias, the emission emanating from the defect site showed single-electron to single-photon conversion. Defect emission with high-purity single photons and femtosecond lifetimes was observed. In another study, W-replaced Se (W_{Se}) antisite defects in WSe_2 /hBN have shown ultralong (225 ns) lifetimes.²⁸¹ From the above studies, it can be understood

that tuning the right defect site in TMDCs makes them a promising candidate to explore either quantum emitter or quantum memory devices, but more research is desired for deterministic placement of defects and identification of defect types based on the above two proposed applications. Recently, interlayer excitons trapped in Moiré potential formed by twisting TMDC heterobilayers by $\sim 60^\circ$ have also shown SPE (Figure 7G). Interlayer excitons showed linear Zeeman splitting and Stark tunability (40 meV) by applying magnetic and electric fields, respectively. Long exciton lifetime and circular polarizability of interlayer excitons trapped in Moiré potential make them different from strain-induced TMDC SPEs.

So far, we have discussed SPE from all excitonic materials, and now we will compare their $g^2(0)$, exciton lifetime, and operating temperature, as shown in Figure 8. II–IV QDs in

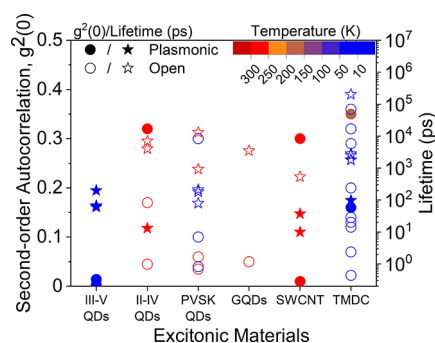


Figure 8. Figure of merit for single-photon emitters. Comparison of second-order autocorrelation function ($g^2(0)$) and lifetime for different excitonic materials: III–V QDs,^{282–284} II–IV QDs,^{259,263} perovskite QDs (PVSQ QDs),^{260,261,266,267,294} graphene QDs (GQDs),²⁶² single-walled carbon nanotube (SWCNT),^{8,9,268} and TMDC.^{269–281} The filled and open circles (or stars) denote $g^2(0)$ (or lifetime) for plasmonic nanocavity and open cavity, respectively. The color coordinate corresponds to the emitter's operating temperature as marked in the color scale.

plasmonic nanocavities have demonstrated room-temperature operation with picosecond lifetimes. Nevertheless, all QDs (II–IV, PVSQ, and GQD) investigated in the open cavity have shown room-temperature operation with long lifetimes (nanoseconds). Such properties may attract QDs for quantum memory devices. SWCNTs emitting in the telecommunication region with high purity and lifetimes close to III–V QDs are a darkhorse, which holds great promise in next-generation communication devices. In contrast to QDs and SWCNTs, emission lifetimes in TMDC SPE can be tuned by varying defect types in TMDCs or placing them in plasmonic nanocavities. Despite extensive work on deterministic placement, scalable patterning, and strain engineering, the TMDC SPE mostly operates at low temperatures (5 K) and shows a relatively large spread in $g^2(0)$, which makes them less viable as a reliable single-photon sources. However, their ability to be integrated with other optical cavities and electrostatic tunability makes them attractive. To raise the operating temperatures and improve $g^2(0)$ confinement strategies such as use of Moiré potential in twisted bilayers and strain-induced confinement are some of the options to consider in future research investigations.

The figures of merit discussed above are critical to compare the best single-photon emitters for a scalable and practical

application. In addition, photon indistinguishability (critical for quantum photonic applications) is also an important metric for any quantum light emitter. Such measurements have been widely performed on more mature semiconductors such as III–V epitaxially grown dots embedded in PhCs and microcavities with 84%²⁹⁵ and near-unity^{296–298} photon indistinguishability. In low-dimensional excitonic systems, there are limited reports of measurements of photon indistinguishability. The low single-photon collection efficiency likely extends the measurement time. This leads to the optical coherence time of the emitter (T_2) to approach twice the spontaneous emission lifetime ($2T_1$), thereby impeding measurement of the photon indistinguishability using the Hong–Ou–Mandel setup.^{261,299} Nonetheless, some reports exist on monolayer TMDC-based indistinguishable photon sources integrated with SiN waveguides and colloidal QDs coupled with nanophotonic resonators that show indistinguishability values of ~ 7 and $\sim 63\%$, respectively.^{299–301} However, none of these values compares well with those of mature entangled photon sources with III–V epitaxially grown QDs, which suggests that excitonic quantum emitters need significantly more progress in both material quality and photonic integration schemes for mature quantum information applications.

The above sections have discussed the fundamentals of light–matter coupling in excitonic semiconductor systems and further detailed the strength and weaknesses of various coupling modalities and semiconductors. In the next section, we will conclude this review with some of the emerging areas of research whose outcome strongly benefits from the light–matter coupling.

EMERGING APPLICATIONS OF COUPLED EXCITONS–PHOTONS IN SOLID-STATE MEDIUM

The electronic band gap of a semiconductor is determined by its crystal structure and its composition. Changing the composition at the atomic scale will tune the electronic band gap. However, using polariton states, the optical density of states in a semiconductor can be tuned based on the cavity design, polariton interaction, and the surrounding dielectric medium. This makes the concept of tuning the optical density of states very powerful for investigations well beyond simple light–matter coupling for emission or absorption enhancement. The polariton lattice is one such concept of constructing an artificial lattice with different optical band gaps using the optical properties of a single semiconductor. These lattices can be constructed in real space using electron-beam lithography,³⁰² ion beam milling,⁶⁶ or laser patterning.³⁰⁶ For instance, perovskite QDs can be filled into nanopillars etched in a DBR to form a perovskite lattice (Figure 9A). Using momentum and real-space imaging, forbidden optical band gaps (13.3 meV) emerging from the upper and lower polariton branches formed as a result of coupling excitons and photons in the real-space lattice can be observed at room temperature (bottom panel, Figure 9A). Similarly, artificial lattices have been extended to other symmetries such as a honeycomb structure (hexagonal symmetry) and a Lieb lattice (square symmetry).³⁰⁷ Real-space lattice engineering to tune the coupling and dispersion of hybrid exciton–photon states can be useful for different applications ranging from realizing flat band Hamiltonians with spin–orbit coupling³⁰⁸ to topological polariton devices³⁰⁹ and even optical simulators³¹⁰ for interacting systems in a nonlinear regime.

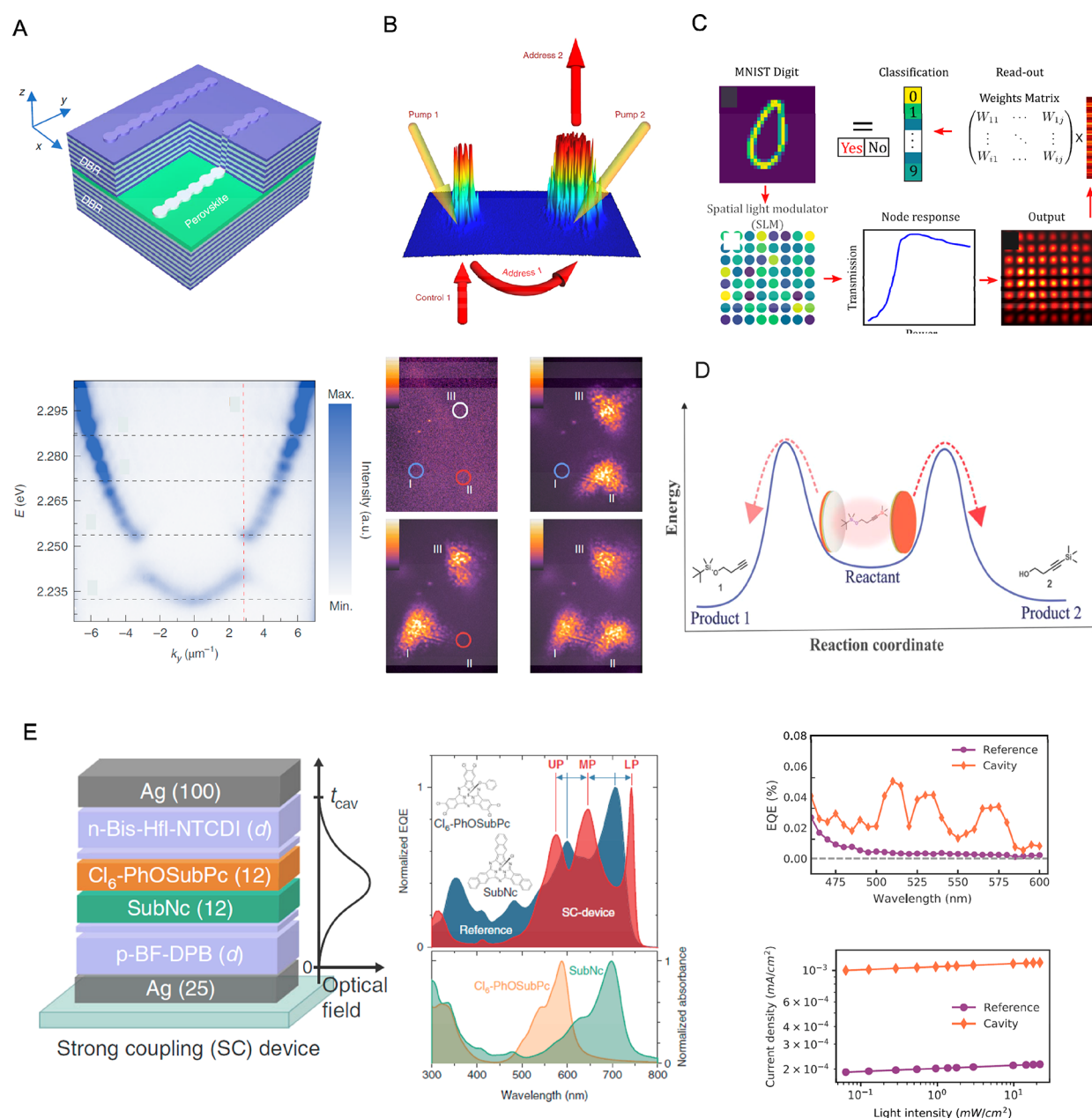


Figure 9. Emerging applications of polaritons. (A) Schematic of one-dimensional CsPbBr₃ perovskite lattice structure sandwiched in a DBR optical cavity to realize polariton lattices (top panel). Experimental momentum–space dispersion showing a forbidden energy gap of 13.3 meV (bottom panel). The black dashed lines represent the energy selections for real-space imaging. The red dashed line represents the edge of the first Brillouin zone. (B) Organic semiconductor-based optical switch with cascaded amplification with two pumps (top panel) demonstrating Boolean OR gate using polaritons. Normalized real-space photoluminescence from three polariton transistors (marked as I, II, and III) comprising the OR gate with four different input configurations shown separately (bottom panel). (C) Schematic illustration of optical neuromorphic computing in In_{0.04}Ga_{0.96}As by exploiting nonlinearities in polaritons. (D) Schematic of a photochemical reaction coordinate modified by placing the reactants in an optical cavity under strong coupling to form polariton states. (E) Schematic diagram of a polariton photovoltaic device made from bilayer organic heterojunctions of donors and acceptors and Ag mirrors on either end forming an optical cavity (left panel). Normalized external quantum efficiency (EQE) from a strongly coupled photovoltaic device compared to the reference device without any strong coupling (middle panel). Enhanced EQE and current density was observed by tailoring the absorption with an optical cavity and absorbing low-energy photons in the lower polariton branch (right panel). Panel A adapted with permission from ref 302. Copyright 2020 Springer Nature. Panel B adapted with permission from ref 69. Copyright 2020 Springer Nature. Panel C reprinted from ref 46. Copyright 2020 American Chemical Society. Panel D adapted with permission from ref 303. Copyright 2019 The American Association for the Advancement of Science (AAAS). Panel E reprinted with permission under a Creative Commons CC BY License from ref 304. Copyright 2021 Springer Nature and Reprinted with permission under a Creative Commons CC BY License from ref 305. Copyright 2019 Springer Nature.

Another tantalizing possibility arising from engineering dispersions of coupled light–matter excitations is to use the frictionless flow of polaritons to exploit them in optical

switches.^{68,69} Polariton states from organic materials strongly coupled to external DBR cavity were used to demonstrate an optical switch by cascade amplification. For this purpose,

excitons dressed with vibrational energy forming “hot excitons” are required to exploit vibron-mediated energy relaxation.⁶⁹ The rigidity in ladder-type polymers offers such a high-energy vibron mode with high oscillator strength. Upon forming polariton states with a DBR cavity, two pumps are used to achieve cascaded amplification in two stages, as shown in Figure 9B. It can be noticed that the condensate emission achieved from pump 1 is further amplified by pump 2. This was used to demonstrate the “OR” binary logic gate, as shown in the bottom panel of Figure 9B. Another Boolean logic gate, “AND”, was also demonstrated in this work. In the following works, two pumps were used as seed pulse and pump pulse. A seed pulse triggers few polaritons, and the nonlinearity and non-equilibrium of the exciton–polariton condensation from organic semiconductors was used to demonstrate a universal NOR and NOT gate by controlling the pump pulse.^{311,312} A subpicosecond, atto-Joule-scale polariton switching and amplification was demonstrated. In addition, this work reports all of these observations at room temperature, in contrast to III–V semiconductor-based optical switches which operate at cryogenic temperatures. Although these recent advancements show that the community is progressing toward realizing efficient optical switches, several challenges such as scalability, energy efficiency, and enhanced speed remain to be addressed.³¹³ Furthermore, the nonlinearity of polaritons can be used in complex pattern recognition which underlies the basis of most neuromorphic computing platforms. To create artificial nonlinear optical neuron III–V semiconductor, quantum wells forming polaritons in a DBR cavity at low temperatures were used.⁴⁶ Handwritten digits were converted into a matrix that is readable by the nodes in the polariton network. Using a spatial light modulator, the digits were encoded into a laser pulse with controllable intensity and phase. Passing this digit-encoded laser into the polariton cavity detected the digits in the transmitted pattern with outstanding accuracy as compared to linear classifiers (Figure 9C).

Hybrid states can also be used to modify chemical reactions by embedding the reactants in a cavity (Figure 9D).³⁰³ Vibrational strong coupling was achieved by vibrational modes from organics such as silane (liquid reactant) resonantly interacting with the cavity modes. Two different products can be formed based on silane cleavage occurring through Si–C or Si–O bonds. Using infrared spectroscopy, the products emerging under strong coupling were detected with high accuracy. Strong splitting of infrared peaks corresponding to Si–C or Si–O bonds were observed by tuning the cavity mode. This technique can be an additional experimental knob to select and accelerate the desired chemical reaction, thereby tilting the chemical landscape using the light–matter coupling phenomenon and in light-driven catalytic chemistry.

TMDC heterostructures placed in an optically resonant structure—either a plasmonic array³¹⁴ or simply on a Au substrate^{315–317}—have shown to enhance the light absorption in the active layer. Such architectures have shown enhanced photocurrent in the photodetectors (or photovoltaics).³¹⁸ Introducing strong light–matter interactions in such exciton–cavity architectures can leverage the potential of photovoltaic devices for generating electricity—“polariton photovoltaics”. Large voltage losses plague the efficiency of organic solar cells during photon conversion. The limiting factor for the energy loss is the difference between optical gap of the solar cells (E_{opt}) and open circuit voltage (V_{OC}) in the photovoltaic devices. To leverage this loss, absorbing sub-band gap photons

can be engineered by introducing strong coupling in an optically resonant structure. Organic heterojunctions placed between two silver mirrors lead to confinement of optical field in the active layer (Figure 9E).³⁰⁵ Consequently, the strong light–matter interaction leads to the formation of hybrid states (UP, upper polariton; MP, middle polariton; LP, lower polariton), which were confirmed from the external quantum yield measurements, and the $E_{\text{opt}} - qV_{\text{OC}}$ was reduced by 60–100 meV. Bulk heterojunctions in strong coupling have also shown 2–3-fold increase in the rate of energy transfer, which in turn enhanced the external quantum efficiency and current density of the device (Figure 9E, right panel).³⁰⁴ Recently, mixtures of two organic J-aggregates when placed in a microcavity have also shown mesoscopic distances of polariton migration within the device. The energy transfer efficiency between two J-aggregates was ~60–80% due to polariton-mediated transfer from the UP to LP through a MP state.³¹⁹ From the above discussion, it is evident that polariton states can result in long-range, coherent energy transfer owing to low effective mass (typically $10^{-4}m_0$, where m_0 is free electron mass) and higher group velocity ($2 \times 10^8 \text{ cm s}^{-1}$) benefiting from the photon component, in contrast to the excitons.³²⁰ Further, the cavity Q factors from the active layer thickness leads to longer photon round-trip propagation time.³²¹ As excitons are known to have limited diffusion length (~10–100 nm) in the organics, introducing polaritons by strong light–matter coupling in excitonic materials-based photovoltaics (for example, organics, TMDCs, and perovskite semiconductors) has the potential for major conceptual as well as performance advance for these class of devices.

CONCLUSION AND OUTLOOK

In this paper, we have reviewed the role of exciton–photon interactions in low-dimensional materials systems ranging from their basic science to measurement to applications. Properties and features of hybrid exciton–polariton states from strong coupling of ensemble particles and Purcell-enhanced emission of single-particle excitations from weak coupling are delineated for these materials.

Over the past two decades or so, tremendous progress has been achieved by the scientific community in not only producing and isolating low-dimensional excitonic materials but also developing measurement and fabrication techniques to resonantly couple photons with these materials. While the use of dielectric Bragg mirrors and plasmonic antennas are more mainstream, the use of extreme dielectric constants of the excitonic materials themselves to induce lossy F-P-like modes in subwavelength dimension semiconductors is an interesting avenue for further exploration. Given the simplicity of the concept and geometry of resulting samples a lot more activity is expected in this area in the future including in the space of photochemistry and photocatalysis. Routes to simplifying the fabrication of electrically controlled devices (for both strong and weak coupling) and enabling direct access to the semiconductor interface for polariton photochemistry are the major challenges when using external optical cavities. This challenge can be addressed by using self-hybridized polaritons as mentioned here. In Figure 2B, we have shown that self-hybridized polaritons (from perovskites, 1D SWCNTs; 3D TMDC multilayer) show high Rabi splitting (marked with arrows), compared to that of their counterparts embedded in an external optical cavity. We believe this insight will motivate

further research toward cavity-less polaritons and address other challenges as discussed in the outlook section.

Another interesting observation that is evident from the literature is the relative scarcity of studying exciton–photon interactions with electrical injection of carriers to produce photons directly inside the cavity. There are many possible reasons for this including lack of appropriate, low-resistance contacting schemes to incompatibility of certain excitonic materials to micro/nanofabrication processes. However, this area does need more attention and development since electrical injection is the likely path to widespread, practical exciton photonic devices ranging from low-threshold polaritonic lasers to electro-optic modulators and switches.

Further, given the maturity of the field of quantum optics and quantum information theory, it is imperative that excitonic materials will be looked at as possible candidates for miniature, integrated, solid-state quantum light sources. In this regard, III–V QDs are the current standard bearers, but the lack of emission in telecom wavelengths is a likely impediment. The key requirements for ideal quantum photon source are many-fold ranging from high brightness to tunability, ease of integration to deterministic positioning, low $g^2(0)$ to near-unity indistinguishability as well as alignment with telecom spectrum. Meeting all of these in one material system is challenging and likely to remain one for some time. Toward that end, 1D excitonic materials like carbon nanotubes and even defect/strain-induced quantum emitters in 2D materials are viable options and must therefore be investigated with more vigor and creative ideas, particularly with regards to scaling up, adjusting emission wavelength in telecom range, and deterministic positioning. Likewise, improving indistinguishability is critical for any quantum applications which will require concurrent improvement in material quality and coupling of the emitters with the optical cavity medium.

Finally, in terms of techniques and sample/materials development, there are several major points worth noting and acting upon. First of all, given the short lifetime of excitons in most discussed semiconductors, it is imperative to develop near- and far-field, real-space transient imaging techniques for coupled exciton photons. While such techniques already exist in the far-field for excitons, time-resolved near-field techniques for excitons that normally lie in visible part of the spectrum are rare. Integrating time- and space-resolved optical measurements in one setup will advance our understanding of exciton properties and additional functionalities at the nanoscale. Replacing the laser source from continuous wave to pulsed laser to acquire time-resolved photoluminescence measurements in the near-field spectroscopy setup can be a first step in this regard. For instance, a combination of near-field and time-resolved measurements investigated in polymer blends were able to identify the competing process between energy transfer routes and trap states.^{322–324} Similarly, time-resolved, scattering measurements using near-field optical microscopy have been performed on TMDCs to visualize the electric field profiles of waveguide modes, which led to observation of a substantial change in the phase velocity in WSe₂ optical waveguides and Stark shift in the A-exciton.³²⁵ Embarking in this research direction will provide additional insights for the quantum optics, nanophotonics, and metamaterials research communities. Further, nonlinear optical processes such as two-photon absorption and second harmonic generation will improve nanoscale imaging resolution by selecting only the optical ranges of interest. Nonlinear two-photon emission

process has been used to study surface plasmon propagation and polaritons in a time-resolved photoemission electron microscopy (PEEM) experiment. Using PEEM, the Meron-like quasiparticles formed by interaction between SPP and spin–orbit interaction,²⁸⁷ as well as exciton–polariton in a perovskite microcavity²⁸⁸ have been directly visualized. PEEM has the advantage of concurrently having high spatial (in nanometers) and time resolution (in femtoseconds) for imaging, which is beneficial for studying exciton dynamics in emerging systems such as Moiré excitons²⁸⁹ and superlattices. Beyond optical response in the visible range, additional information from the material can be identified from the infrared region using a near-field technique. Recently, infrared vibrational spectroscopic studies have shown label-free chemical information and cation vibrational dynamics in liquids³²⁶ and perovskites,³²⁷ respectively. Extending the laser excitation up to the ultraviolet range in near-field spectroscopy studies will help characterizing low-dimensional semiconductors such as perovskite QDs and its superlattice,³²⁸ organics on 2D hybrids,³²⁹ and biological materials which have shown extraordinary optical properties in the far-field. Identifying tip materials and modified optics that can resonate at ultraviolet frequencies will also be critical to realizing this step.

Overall, the past two decades have witnessed tremendous progress in excitonic materials and optical techniques to couple photons strongly with them and characterize them. The coming decades are expected to take this field which has largely been in the domain to basic science into a domain to practical technology with improved materials, interfaces, and optical cavities. Likewise, on the basic science front, it is expected that improvements in imaging and spectroscopic techniques will shine more light on the transient nature and dynamics of coupled exciton–photons and gain attraction toward fundamental research in both condensed matter physics, chemistry, and quantum information sciences.

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Author Contributions

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Notes

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